

Chapter 167

Resolve FX Texture

These plugins are designed to add texture to images, to replicate both naturalistic and other effects.

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Analog Damage (Studio Version Only)

A Texture category plugin that simulates different kinds of signal degradation resulting from analog transmission and recording, Analog Damage can be used to create “old TV” or “junky videotape” effects of various kinds. A Preset pop-up menu lets you choose from a variety of different looks, while a Custom option lets you create your own.



(Top) The original image, (Bottom) The image with Analog Damage applied

Preset: Lets you add a number of video damage types to your footage that DaVinci Resolve was designed to repair in the first place.

It has the following categories of parameters:

Telecine Source: Options for lens vignetting and automated shutter weave to the left and right.

- **Vignetting:** Darkens the image around the edges to replicate a poor lamp in a telecine.
- **Vignette Aspect:** Adjusts the width of the vignette effect.
- **Shutter Weave:** Replicates the slight movement of the image as the film reel unwinds on a telecine.

Broadcast Signal: Options for customizable noise, detail loss, signal ghosting, and chroma misalignment.

- **Noise Scale:** Determines how fine the noise is.
- **Signal Noise:** Determines how much static is in the Luma signal.
- **Chroma Noise:** Determines how much static is in the Chroma signal.
- **Detail Loss:** Determines how much of the detail is lost in “transmission.”
- **Chroma Detail Loss:** Determines how much of the chroma detail is lost in “transmission.”
- **Ghosting:** Replicates the ghost image due to a reflected signal.
- **Ghost Offset:** Lets you reposition the ghost image.
- **Chroma Misalignment:** Determines how offset the color signal is from the brightness.

Color Dials: Television-signal specific color adjustment controls.

- **Brightness:** Replicates a consumer television brightness control knob.
- **Contrast:** Replicates a consumer television contrast control knob.
- **Color:** Replicates a consumer television color (saturation) control knob.
- **Tint:** Replicates a consumer television tint control knob.

Scan: Controls to simulate problems with television scanning.

- **Image Aspect:** Adjusts the shape of the frame, padding or cropping as necessary.
- **H-Shift:** Lets you roll the image horizontally.
- **V-Shift:** Lets you roll the image vertically.
- **V-Hold/Latch:** V-Hold interacts with the V-Hold Latch parameter to make it easy to use simple key framing to trigger animated vertical rolls that “bounce” over time in a realistic way for broadcast.
- **Overscan:** Allows you to adjust how much of the image is cropped for display on the screen.
- **V-Scale:** Lets you adjust the vertical scaling of the image.
- **Vertical Blanking:** Lets you determine the Vertical Blanking Interval size.

Scan lines: Simulate customizable television scan lines that are capable of simulating moire and color artifacts.

- **Line Sharpness:** Lets you adjust the visible sharpness of the scan lines.
- **Line Frequency:** Lets you adjust how many scan lines are in the image.
- **Colored Lines:** Check this box to add color outlines to the scan lines.

TV Construction: Controls to simulate issues with CRT phosphor brightness, tint, and defocus, and the ability to add a customizable curved border and warped screen curvature.

- **Phosphor Brightness:** Lets you adjust the brightness of the front screen, even when the tv should be “off.”
- **Phosphor Tint:** Lets you adjust the color of the front screen, even when the tv should be “off.”
- **Defocus:** Lets you adjust how much detail is lost to the screen.
- **Screen Curvature:** Lets you adjust how curved the picture tube is.
- **Edge Mask:** Check this box to add a black mask around the edge of the screen.
- **Edges Transparent:** Check this box to make the screen edges transparent for compositing.
- **Mask Curvature:** Lets you adjust the shape of the black mask.
- **Mask Aspect:** Lets you adjust the black mask’s aspect ratio.

VHS: Lets you add a customizable “Restless Foot” to simulate recorded tape artifacts.

- **Restless Foot Height:** Lets you set how much restless foot intrudes from the bottom of the screen.
- **Restless Foot Offset:** Lets you choose which direction and how much to distort the restless foot.
- **Restless Foot Jitter:** Lets you adjust the shaking of the restless foot between frames.

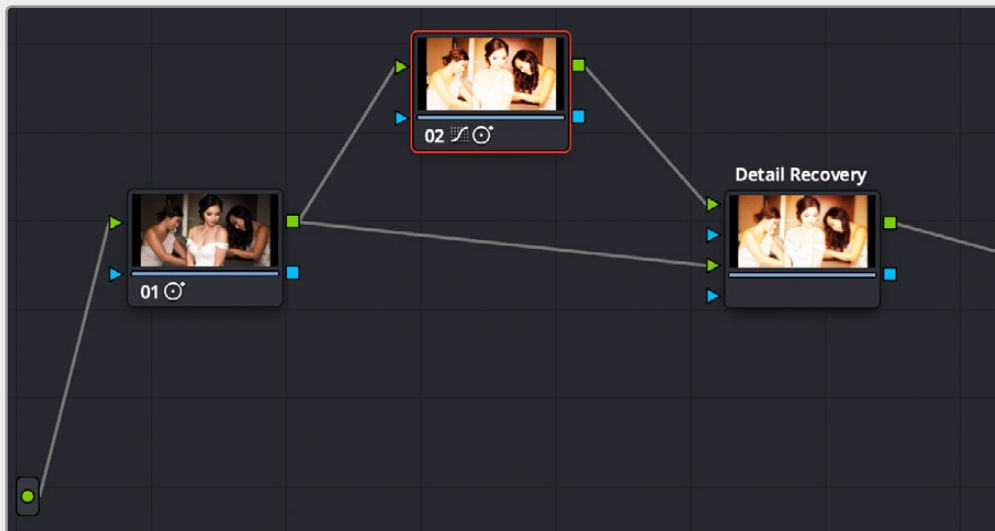
TIP: Using the Analog Damage plugin on the Fusion page makes it possible connect Modifiers to add automatic animation (such as Perturb) to different parameters in order to quickly and easily create automated video damage effects.

Detail Recovery (Studio Version Only)

Detail Recovery is a utilitarian effect that lets you extract image detail from the second input, in order to re-add it to the image coming in the first input. In this way, you can selectively add detail back to an image in which it's been removed. Because of how it works, this is an effect that's intended to be used on the Color page.

To use Detail Recovery, you must add it to the node tree as a Resolve FX node (dragging it directly from the Open FX library into the node tree as its own node), in order to expose both of the RGB inputs that appear on the node when you do this. By default, the first RGB input is for the image you want to add detail to, and the second RGB input is for the image you're extracting image detail from.

In the following example, Node 1 is the base grade, and Node 2 is a grade in which the highlights are being blown out for stylistic effect. The Detail Recovery node is then being added to add specific image detail from Node 1 back to the output of Node 2.



Setting up the Detail Recovery effect to add image detail from Node 1 to the image coming from Node 2

The result, compared with the image in Node 2, is an image that's still harshly exposed, but that has traces of fine image detail so it doesn't look so clipped.



(Top) The image with highlights compressed harshly, (Bottom) The image with high-frequency detail added back

Detail Recovery has the following parameters.

Detail Recovery

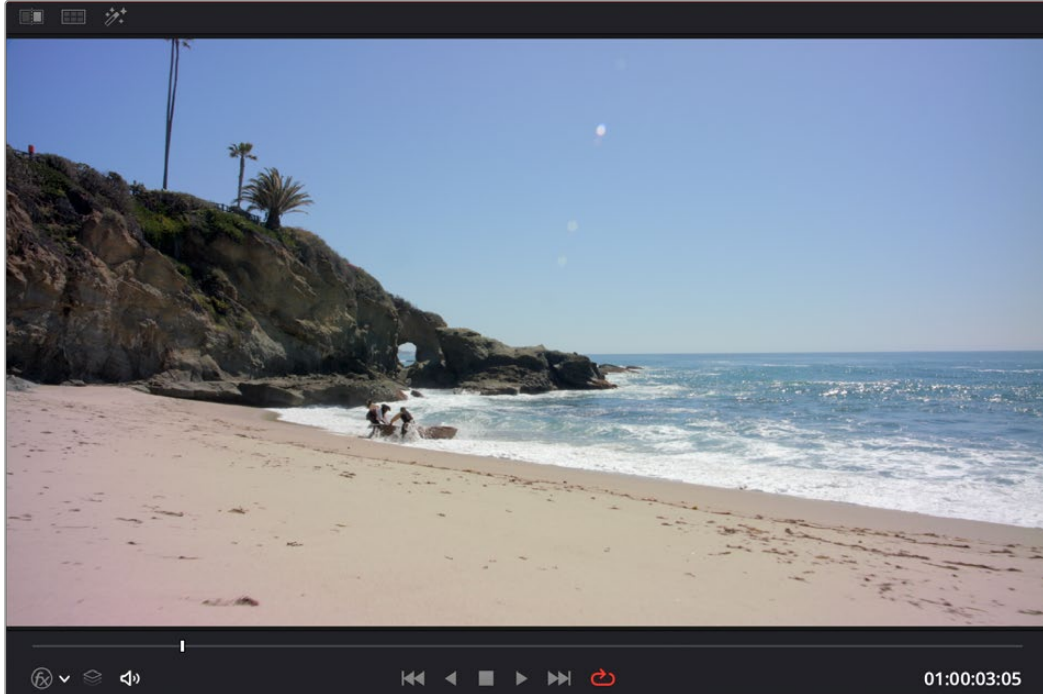
- **Transfer Direction:** This drop-down menu lets you choose which input provides the detail and which input the detail is combined with.

Details Extraction

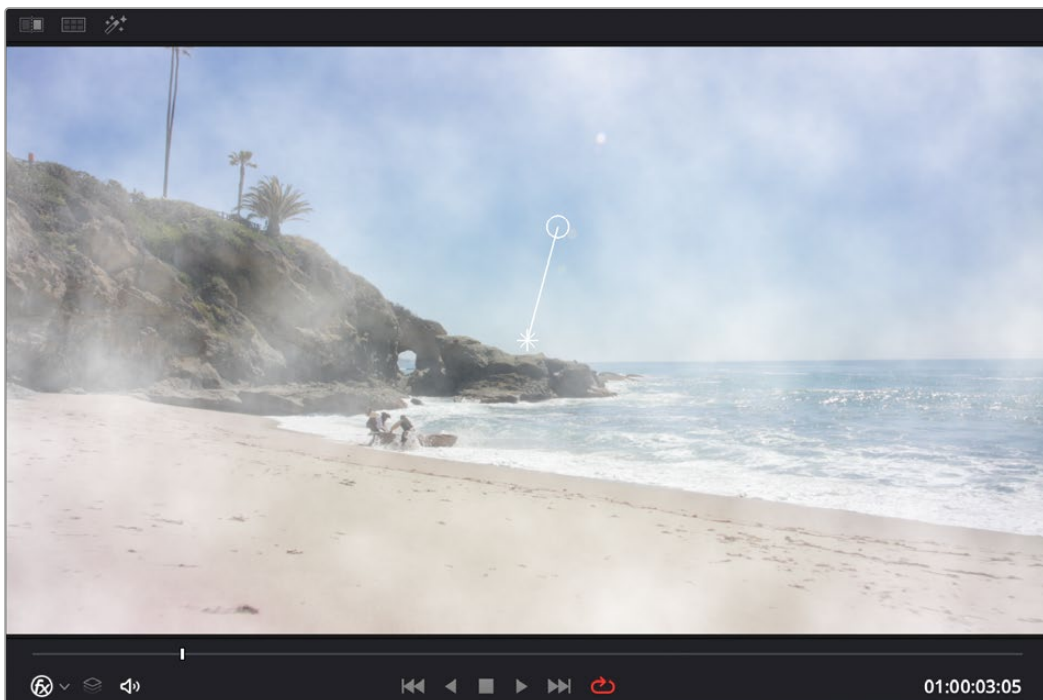
- **Frequency Cutoff:** A slider that lets you choose how much detail you want to extract for recovery. Lower values isolate finer details; higher values include progressively larger details.
- **Detail/Edge Balance:** This lets you choose between recovering all details or from the stronger edges only.
- **Strength:** This slider lets you exaggerate or attenuate the detail that's extracted for recovery.
- **Detail Mix:** At a value of 1.000, this slider blends the extracted detail with the image you're adding it to so that both layers blend together. At a value of 0.000, the extracted detail replaces whatever overlapping pixels are in the image you're adding it to. Using this slider, you can choose the best blend for your purposes.
- **Preview Detail checkbox:** Turning this checkbox on lets you see the detail that's being extracted using the Frequency Cutoff and Strength sliders.

Fast Noise

Fast Noise creates computer-generated noise patterns that can be composited on, or used to warp a video clip for a variety of special effects, such as smoke, haze, or water ripples.



Original image



The image with Fast Noise's Smoke preset applied.
The onscreen controls for position and rotation are visible.

Appearance

Controls for what noise pattern is generated. All parameters are keyframeable for changing the noise pattern over time.

- **Preset:** Select from some common noise pattern use cases: Mist, Smoke, Water Surface, River, and Heat Haze. Default is the standard noise pattern, and Custom is auto-selected if you change any of the parameters below.
- **Scale:** The size of the noise pattern.
- **H/V Ratio:** Controls the horizontal/vertical spread ratio.
- **Detail Level:** How many levels of fine detail to generate.
- **Detail Balance:** Applies weighting to the different scales of detail. The slider lets you choose between stronger large detail or stronger fine detail. A value of 0.000 removes fine detail entirely.
- **Evolution:** Changes the noise pattern gradually. This slider controls the “progress” of the noise pattern over time, rather than just modifying its current state.

Adjustment

Controls for adjusting the parameters of the noise pattern once it's been generated. Especially useful for modifying the noise pattern's Alpha channel for compositing. All parameters are keyframeable for changing the noise pattern over time.

- **Brightness:** Increases or decreases the gain of the noise pattern.
- **Contrast:** Moving this slider to the right makes the light areas lighter and dark areas darker in the noise pattern, increasing the contrast. Moving it to the left reduces the contrast.
- **Saturation:** This slider controls the scale between a grayscale and color noise pattern. A value of 0.000 is completely grayscale, and a value of 1.000 is fully color.
- **Color Tint:** Sets the overall color of the effect.
- **Invert Dark Regions:** Check this box to invert the dark regions to white and darken the edges in the noise pattern.
- **Invert Pattern:** Check this box to invert the color of the noise.
- **Allow Negative Values:** Check this box to apply negative values to the Output section below.

Position

Controls that adjust the center and direction of the noise pattern.

- **Position X/Y:** Location of the center of the noise pattern.
- **Rotation:** Rotates the noise pattern around its center.
- **Position Reference:** Controls how to position the effect when a clip is re-sized or moved in the Timeline.

Sizing and FX Tracker: The noise pattern follows Sizing and FX Tracker adjustments in the Color page. For example, if you used the FX Tracker to track a camera pan, the noise pattern would move with the pan, rather than stay in one place as the shot panned underneath it.

Keyframed Positions: The noise pattern position can be adjusted as normal using keyframes.

The position and rotation controls can also be adjusted directly in the Viewer when the Open FX Overlay mode is selected from the drop-down menu in the bottom left of the Viewer. Dragging the star in the Viewer adjusts the X/Y position. Dragging the circle clockwise or counterclockwise sets the rotation.

Auto-Animation

These controls allow you to adjust the animation of the noise pattern.

- **Velocity X/Y:** How far to automatically advance each frame from the start position. You can keyframe these controls to change the noise pattern's direction, without adjusting its actual X/Y position. This lets you mimic motions like smoke changing direction in the wind.
- **Seethe:** Controls how far to automatically evolve each frame from the starting Evolution value.
- **Randomize Start Frame:** Aligns the skipping cycle to the current frame. This allows you to re-use a favorite effect in multiple clips without it looking exactly identical in each instance.
- **Start Frame:** Lets you set the reference frame for the skip pattern. This allows you to control the sequence of noise you get, just in case you need to match it to the same noise pattern on another clip. This frame will be repeated.

Output

These controls allow you to choose how to adjust the output of the noise pattern.

- **Preview Noise:** Checking this box shows the noise pattern alone.
- **Output:** Choose how to use the noise pattern.

Composite Onto Clip: The noise pattern is placed on top of or mixed into the video clip.

Composite Type: Selects composite mode for the noise pattern.

Put Into Alpha Channel: The noise is inserted into the node's Alpha channel, allowing you to use the noise pattern for compositing purposes. Please note, you will also need to right click on the node and select "Use OFX Alpha" to activate this setting. Alternatively, you can drag the Fast Noise effect directly into the Node Graph as an FX Node instead of applying it to a corrector node, and you can use the Alpha without additional settings.

Use to Warp Image: The noise pattern is used to distort the image.

Input Alpha: Controls what to do with the input Alpha channel.

- **Limits Warping Effect:** Limits the warp to certain areas as defined by the Alpha channel. For example, when you want to have a water ripple or heat shimmer on only certain part of the clip.
- **Gets Warped as Well:** The warp affects the Alpha channel in addition to the RGB channels. For example, when warping a logo with a transparent background.

JPEG Damage

Simulates the kinds of minor or major artifacts you get when doing JPEG compression; useful for simulating compression damage.

- **Quality:** Lets you lower the bit depth of the image.
- **Resolution:** Lets you increase the size of macroblocking artifacts that appear.
- **Block Aspect Ratio:** Lets you adjust the macroblocks to be more square or rectangular.
- **Frequency Scale:** Sharpens the effect.
- **Scale Component:** Lets you base this effect on All Frequencies, the X-Frequency, or the Y-Frequency.

Texture Pop (Studio Version Only)

Texture Pop is a more sophisticated and flexible version of the Midtone Detail control found in the Primaries and HDR palette. Using the controls of this effect, you can either remove texture or exaggerate it. There are two modes of operation that determine how specific a result you can achieve.

- In Simple mode, you can apply more extreme versions of the Midtone Detail effect found in the Primaries and HDR palette. A single Details slider lets you soften or sharpen midtone detail in the image with a greater range of operation, while a Strength slider lets you exaggerate the effect further or attenuate the effect to back it off.
- In Advanced mode, you can fine-tune the softness or sharpness of high, medium, and low-frequency image detail with great specificity via seven individual “details” sliders, labeled Rough, Coarse, Medium, Small, Fine, and Tiny. This lets you adjust the textural characteristics of multiple frequencies of image detail in different ways, for example softening Fine detail while intensifying Medium detail. The Strength slider lets you exaggerate or attenuate the overall effect caused by all sliders.

In either mode, the Details/Range sliders default to 0.000, which causes no change to the image. This is the level to choose to preserve the original quality of detail in the image. They range from -1.000 (maximum softening) to +1.000 (maximum sharpening). Additionally, a set of Tonal Range controls (described later) lets you choose how much of this effect is applied to the Shadows, Midtones, and Highlights of the image, giving you even more specific control over which parts of the image are softened and/or sharpened.

Understanding Advanced Mode

Advanced mode is the most exciting way to use this effect. The best way to learn what Advanced mode does is to set all of the Details sliders to -1.000, and then raise each one individually to see what kind of image detail get reintroduced, and ultimately enhanced, if you raise each slider above 0.000. In the following image, each of the Details sliders has been set to -1, which is the maximum smoothing for each setting. The result is an image with minimal detail.

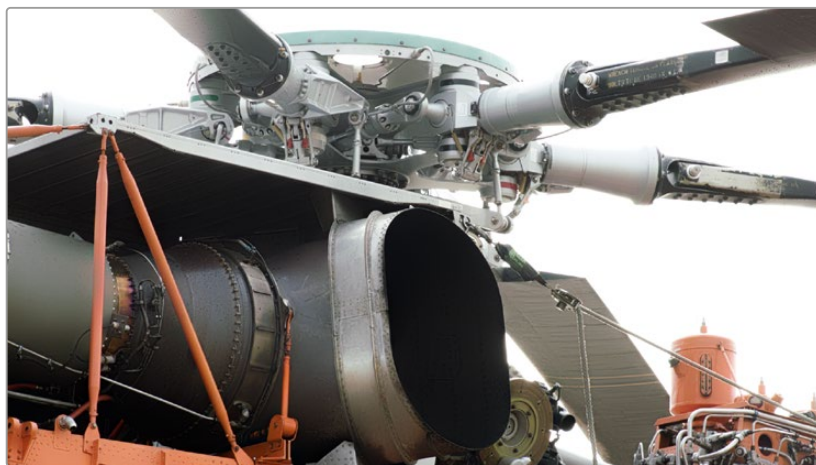
In the following example, a person and the mechanical details of a helicopter shot are smoothed by setting each detail slider to -1.000. You can see that the resulting images both retain the broad strokes of color and contrast that define each region of color, but outside of high-contrast edges, all textural detail has been smoothed to a clean gradient.





(Previous page) An image of a person with all detail smoothed,
(Above) An image of mechanical detail with all detail smoothed

Then, by raising each of the Details sliders back to 0 or even +1 one at a time, you can see exactly what kind of detail that slider is affecting. As you can see in the example of the person, the Tiny and Fine sliders are affecting only the smallest details, hairs, lines, and pores of the skin. In the helicopter detail, you can see the kind of small textures, tiny lines, and rivet details that come back.



(Top) The same image of the person with the Tiny and Fine sliders raised to add subtle detail back to the image, (Bottom) The helicopter detail with Tiny and Fine detail added back to the image

Here are two sets of generalizations that hold true if you're thinking about the detail of a human face.

- **Tiny, Fine, and Small** controls affect the smallest details in an image, including the pores (tiny), blemishes (fine), and wrinkles (small) of average closeup shots. Other features affected would include high-frequency edge detail of strands of hair, leaves on a tree, or text on a page of paper or screen.
- **Rough, Coarse, and Medium** tend to affect progressively smaller shadow contouring of faces, bodies, and objects.

Of course, the granularity of a particular type of image detail depends on how much of a closeup or long shot the clip is. When the camera is closer to the subject being adjusted, you'll be able to adjust much finer details. When the camera is distant, then even large details on the subject are affected by the Tiny, Fine, and Small controls. This means when you create an adjustment for a subject in a closeup, you'll need to make changes to the adjustment if you try and apply it to a long shot.

Mode Controls

- **Operating Mode:** Lets you choose between the Simple and Advanced modes of this plugin.
- **Color Mode:** Lets you choose between working in RGB and Luma/Chroma mode. In RGB, a single set of parameters affect all three color channels at once. In Luma/Chroma mode, two sets of controls let you fine-tune the luma and the chroma separately.
- **Output Mode:** Lets you choose between outputting the Final Result, Differences mode, which lets you see the effect differenced against the original image, and Difference Magnitude, which shows a high-contrast differencing comparison.

Details Controls

- **Strength:** Lets you exaggerate the effect of the other Details sliders further, or attenuate the effect to back it off.
- **Details:** (Simple mode only) Lets you soften or sharpen midtone detail in the image with a greater range of operation.
- **Rough/Coarse/Medium:** (Advanced mode only) These three sliders affect progressively larger contours of the image, appearing to affect the shadows and structure of the image.
- **Small/Fine/Tiny:** (Advanced mode only) These three sliders affect progressively smaller details and texture of the image, adjusting medium to tiny details, lines, and specks.

Tonal Range Controls

- **Shadows:** Lets you reduce the effect you've created in the shadows of the image, as defined by the Low Range parameter.
- **Midtones:** Lets you reduce the effect you've created in the midtones of the image, as defined by the Low and High Range parameters.
- **Highlights:** Lets you reduce the effect you've created in the highlights of the image, as defined by the High Range parameter.
- **Low Range:** Lets you define the image value that defines the transition from Shadows to Midtones.
- **High Range:** Lets you define the image value that defines the transition from Midtones to Highlights.